

Responding to News Sensitively in Stock Attention Networks via Prompt-Adaptive Trimodal Model

Haotian Liu, Bowen Hu^{ID}, *Graduate Student Member, IEEE*, Yadong Zhou^{ID}, *Member, IEEE*, and Yuxun Zhou^{ID}

Abstract—Modern quantitative finance and portfolio-based investment hinge on multimedia news and historical price trends for stock movement prediction. However, prior studies overlook the long tail effect in the feature distribution of stocks, inevitably leading to biased attention and thus degrading the efficiency of utilizing news. To this end, we propose a prompt-adaptive trimodal model (PA-TMM) to overcome the biased stock attention networks and tail feature scarcity problem. In this model, sentiments automatically extracted from trimodal information serve as prompts reflecting the market’s collective mood for other entities, and the interactions among stocks are dynamically inferred for integrating both news- and price-induced movements. By leveraging the movement prompt adaptation (MPA) strategy, our model proactively adapts to the feature-imbalanced phenomenon and converges toward being responsive to the news sensitively. Extensive experiments conducted on real-world datasets consistently demonstrate not only the superiority of the proposed framework over various state-of-the-art baselines, but also its effectiveness, profitability, and robustness in Fintech. The code is accessible at <https://github.com/lauht/PA-TMM>.

Index Terms—Graph attention networks, long-tailed feature distribution, stock movement prediction, trimodal fusion.

I. INTRODUCTION

STOCK markets, which constitute a dynamic real-world financial ecosystem, play an indispensable role in driving economic development and facilitating diversified investment opportunities. Nowadays, the rapid growth of multimedia platforms, including news outlets and social media, has revolutionized how stakeholders engage in comprehensive evaluations to identify stocks with the most investment potential, which further influences various financial activities such as asset allocation, risk management, and hedging [1], [2], [3], [4], [5].

In real-world investment, even the subtle insights from financial news can be parlayed into significant profit indicators. Thus, researchers have long been drawn to the idea of gathering information from the news or social media to augment their analysis [6], [7], [8], [9], [10]. While stock price signals inherently contain randomness, research in this field specifically

focuses on predicting the deterministic component brought about by news, rather than the random component. Typically, each stock has two other essential feature modalities for financial time-series forecasting: multivariate time-series features such as opening price, closing price, and volume, and discrete tabular features, i.e., technical indicators calculated based on historical trading signals in the first category. However, an underlying assumption in these studies is that trading signals for all stocks are mutually exclusive [11]. Since all stocks belong to the same financial system, each stock is inevitably affected by peer stocks due to factor movement, lead-lag effect, portfolio rebalancing, etc. [12], [13], [14], which is referred to as momentum spillover effect [15]. Recently, with the advent of graph neural networks (GNNs) [16], [17], more and more studies have conceptualized the stock market as a complex network to model peer interactions, wherein each entity is treated as a node, and the edge between two nodes represents their relationship [18], [19], [20], [21], [22]. These relations are often modeled by the hard-coded microstructure such as time-series correlation [3], supply chain [23], and news co-occurrence [24]. Although these methods capture the interactions to some extent, the static network structure still struggles to adapt to the dynamics of the real-world financial market. Some existing methods based on graph attention networks (GATs) [25], [26] attempt to dynamically assign weights (attention scores) to neighbors to model adaptive interactions between stocks as stock attention networks [6], [27].

However, real-world financial data for algorithmic trading often exhibit a long-tailed feature distribution on each trading day. As illustrated in Fig. 1 (top), only time-series features and tabular indicators (known as head features) for all stocks are available, while the news (known as tail features) dynamically covers only a fraction of stocks. As a result, when introducing external information in GAT-based methods [6], [15], investment insights from news may be overwhelmed by massive price-related head features. Specifically, most graph attention mechanisms are supported by the similarity between two node representations (queries and keys) [6], [15], [25], [26], which leads to a biased attention effect since one stock is easily distracted by the abundance of trading signals and peer stocks, as exemplified in Fig. 1 (bottom). In this case, breaking news from the financial sector, especially from banks, will inevitably drive market sentiments and further impact the overall stock market, attributed to market anomalies [21], [28], yet they receive insufficient cross-sector attention.

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Haotian Liu, Bowen Hu, and Yadong Zhou are with the Ministry of Education Key Laboratory for Intelligent Networks and Network Security, Faculty of Electronic and Information Engineering, Xi’an Jiaotong University, Xi’an, Shaanxi 710049, China (e-mail: ydzhou@xjtu.edu.cn).

Yuxun Zhou is with the Department of Electrical Engineering and Computer Sciences, University of California at Berkeley, Berkeley, CA 94720 USA.

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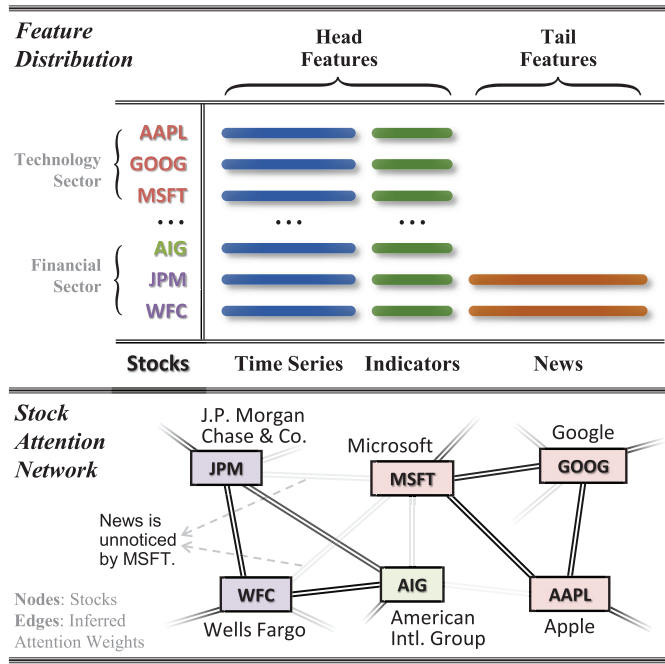


Fig. 1. Example of the long-tailed feature distribution of stocks causing biased attention for MSFT on a certain day, where breaking news from financial institutions (JPM and WFC) should have been crucial for any stock.

Different from long-tail learning focusing on the class-imbalanced phenomenon [29], [30], [31], overcoming the long tail effect in feature distribution in GAT-based stock prediction tasks presents us with two main challenges. One is the feature-imbalanced problem, which causes the attention mechanism to bias toward the dominated head features, and another is the data scarcity problem leading to the poor generalization on those rare tail features, i.e., media news. An intuitive approach is resampling [32], [33], [34], i.e., creating the replicate of the dataset to optimize all model parameters. However, existing resampling techniques are initially proposed for the class-imbalanced problem and thus do not apply to financial scenarios with feature imbalance. Simply imitating their strategies, e.g., resampling past daily news from the same day, might not prove effective as it does not ensure that the model adapts to the diverse and intricate future market landscapes.

Fortunately, one idiosyncratic nature of financial news brings about a turning point. Indeed, when events emerge that affect specific stocks (or companies), breaking news often exerts an instantaneous dominance over their movements. For example, in January 2024, the Federal Aviation Administration ordered airlines to ground more than 170 Boeing 737 aircraft. Following this directive, Boeing's stock closed with a drop exceeding 8% and the trading volume surpassed ten times that of the previous trading day. Meanwhile, Airbus, Boeing's primary competitor, experienced a slight increase in stock price. This implies that news information does not hold the same priority as price information, but few studies have delved into this characteristic and incorporated it into the model.

Such dominance of news establishes a near-equivalence between news and stock price movements. Then, it is possible

to design a resampling strategy that bypasses news and directly generates large amounts of putative news sentiments (treated as news) from stock price movements (labels) for data augmentation. This helps the graph attention mechanism to be pretrained to adapt to more complex and extreme news coverage, thereby improving the model's generalization performance and addressing the long-tail feature distributions overlooked by existing methods. However, realizing this concept requires a redesign of the model architecture and the training strategy so that the attention mechanism can simultaneously accommodate both news information and the generated putative news. It also requires the model to transform the news into potential stock movements (or news sentiments) before calculating the graph attention weights to satisfy a seamless integration of pretraining and fine-tuning.

In this work, we propose a prompt-adaptive trimodal model (PA-TMM) for stock movement prediction to address the challenges of long-tailed feature distribution. Specifically, we begin by designing two subnetworks: a *cross-modal fusion* module for integrating trimodal features and extracting the news-induced sentiments as prompts for other stocks, and a *graph dual-attention* module for dynamically inferring the stock attention network by a graph dual-attention mechanism, which circumvents direct similarity measurement of heterogeneous representations generated from news and prices and thus facilitates overcoming biased attention. To tackle the tail feature scarcity problem, we introduce an equivalence resampling (EQSamp) strategy for data augmentation by establishing a direct connection between market sentiments in news and stock movements by considering the dominant impact of news. Furthermore, to facilitate the model in proactively adapting to extreme feature imbalance and converging to be sensitive to tailed news, we pretrain our model using augmented data with generated prompts. Finally, we fine-tune the model with real-world data, mainly focusing on understanding news representations. In summary, our main contributions are listed as follows.

- 1) By incorporating the dominance of news, we propose a learning framework named PA-TMM for stock movement prediction, which achieves heterogeneous modality fusion and effectively captures news propagation dynamics by graph learning.
- 2) For our framework, we adopt a targeted pretraining method named movement prompt adaptation (MPA), which not only navigates the learning system to respond to tailed news sensitively but also prevents it from overfitting due to over-reliance on stocks carrying news.
- 3) We propose an EQSamp strategy for financial data augmentation when pretraining to overcome the news scarcity problem, thereby enhancing the generalization ability of graph attention networks on feature-imbalanced datasets.

The subsequent sections of this article are organized as follows. Section II discusses related works. Section III defines the stock movement prediction problem and the trimodal features used. Section IV presents the PA-TMM architecture. Section V presents the optimization strategy tailored for

PA-TMM. We present the experimental results in Section VI and conclude this article in Section VII.

II. RELATED WORK

A. Time-Series Stock Prediction

In stock movement prediction, traditional time-series learning methodologies involve encoding an individual stock as a sequential latent representation for downstream tasks [35], [36]. They often depend on diverse variants of recurrent neural networks (RNNs), such as long short-term memory (LSTM) [37] and gated recurrent unit (GRU) [38], or Transformers [39] to capture the underlying time-varying patterns from multiple time steps, wherein each stock is represented by its temporal price and volume signals. For example, PEN [40], MAN-SF [12], and MTR-C [41] encode the time series for each stock using RNNs. Some studies were also devoted to mingling different types of market factors [42], investment behaviors [43], and technical indicators to improve prediction performance. For example, relational event-driven stock trend forecasting (REST) [44] utilizes the event information from the company's announcements to further model the stock's reaction to historical events. Digger-Guider [45] can produce powerful high-frequency stock factors that significantly improve stock trend prediction performance. However, an inner motivation in these studies is that the trading signals of all stocks are mutually exclusive. This assumption is not consistent with the highly internally coupled financial markets in the real world due to phenomena like momentum spillover [13], [15].

B. Graph-Based Stock Prediction

Given that all stocks belong to the same trading system, the movement of each entity is inevitably influenced by its peer entities [46]. In particular, ample evidence exists, indicating a lead-lag effect in the stock market, where the stock prices of certain firms manifest a delayed reaction to the price innovations of other firms [47], [48]. To model this intraindustry phenomenon, pilot studies conceptualize the stock market as a graph, with each entity treated as a node, and edges defined by the relations (e.g., industry category [2], supply chain [23], business partnership [6], price correlation [3], lead-lag correlation [2], and causal effect [21]) between two nodes. As a next step, variants of GNNs are employed to aggregate the influences from related stocks. For example, THGNN [3] generates a temporal and heterogeneous graph for graph aggregation operation. ESTIMATE [2], utilizing hypergraphs based on industry classifications, captures nonpairwise correlations among stocks. SAMBA [49] models dependencies between daily stock features by utilizing a bidirectional Mamba block and an adaptive graph convolution module. By aggregating peer influences, these graph-based methods update node representations to capture neighbor-induced movement [46].

C. News-Based Stock Prediction

With the advancement of multimedia technology, fundamental analysis introduces external information beyond the trading market by integrating financial news [40], [50], [51] or social media posts [7], [20], unveiling comprehensive

information that extends well beyond prices. For instance, multi-source aggregated classification (MAC) [1] leverages the graph convolutional networks (GCNs) to aggregate the effects of news on related companies. NumHTML [52] and multi-view fusion network (MFN) [53] aggregate various features such as technical indicators and textual news to glean investment insights. To adapt to the market dynamics, several attempts have been made to model the time-varying structure of stock networks when combining stock interactions and news information. For instance, AD-GAT [15] and DANSMP [6] adjust or infer the edges by different variants of GAT using time series of prices, market sentiments from news, etc. Multi-scale multi-modal fusion (MSMF) [54] balances complementarity and redundancy across modalities by integrating the modality completion encoder, multiscale feature extractor, and fusion mechanism. Nowadays, the methods for stock movement prediction are evolving from a unimodal to a multimodal paradigm, paving the way for advancements in portfolio-based investment strategies. However, due to the lack of consideration for the long-tail effect, most of these methods simply concatenate features from different modalities, making it difficult to fully and effectively leverage news information and model the propagation of news impact within stock networks. In this article, we propose a pretraining strategy for GATs in finance that enables the model to proactively adapt to the imbalanced feature distribution, thereby enhancing the generalizability.

III. PROBLEM STATEMENT

In this section, we provide a formal definition of the stock movement prediction problem. Generally, time-series forecasting (or prediction) is typically regarded as a regression task. However, in the Fintech domain, particularly in the stock market, predicting the exact value of stock prices is far more challenging than predicting price movements. As a result, most research efforts focus on using time series as (a part of) input features to optimize models, with the objective of outputting a class label that indicates the rise or fall of stocks [6], [40], which are determined by comparing whether the stock price on the current trading day is higher than that of the previous trading day. Here, we aim to leverage trimodal features on the $(t - 1)$ th day as input features to predict the movements (labels) for all stocks in the stock pool on the upcoming trading day t . Although the time-series data are utilized, the proposed method is a classification method optimized through an objective function for classification.

Given a set $\mathcal{V} = \{1, \dots, N\}$ of stocks, the movement (labels) y_i^t of stock $i \in \mathcal{V}$ on day t is defined as

$$y_i^t = \begin{cases} 1, & c_i^t > c_i^{t-1} \\ 0, & c_i^t \leq c_i^{t-1} \end{cases} \quad (1)$$

where c_i^t and c_i^{t-1} correspond to the adjusted closing price¹ of stock i on day t and the previous day $t - 1$, respectively.

Formally, we have three feature modalities for prediction: 1) textual news corpora $\mathcal{T}$ that have labeled the relevant stocks impacted by each news item; 2) historical time-series

¹The adjusted closing price of a stock reflects the stock's closing price after accounting for corporate actions and events that may affect the price.

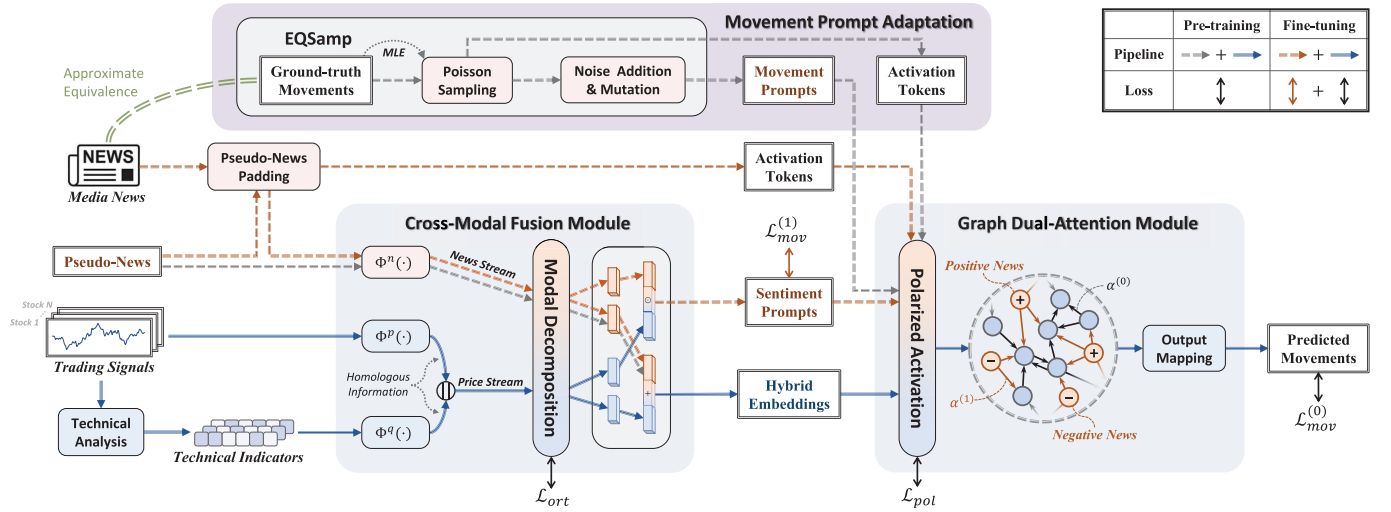


Fig. 2. Framework of the proposed PA-TMM, which comprises two main subnetworks: the *cross-modal fusion* module and the *graph dual-attention* module. A two-stage optimization method is employed to facilitate the model in adapting to feature imbalance proactively and converging to be sensitive to tail features, i.e., media news.

trading signals $\mathbf{X}_i^{[t-T:t]} = [\mathbf{x}_i^{t-T}, \dots, \mathbf{x}_i^{t-1}]$ for each stock $i \in \mathcal{V}$ from past T trading days; and 3) tabular technical indicators $\mathbf{I}_i^{t-1} \in \mathbb{R}^{d_f}$ for each stock $i \in \mathcal{V}$ on the $(t-1)$ th day. Here, $\mathbf{x}_i^\tau \in \mathbb{R}^{d_x}$ is the transaction features of stock i on the τ th day, including the highest and lowest price, opening and closing price, trade volume, and rankings of these values over 5 days, 20 days, and 60 days. To address varying price levels, we standardize the price values of each stock (excluding rank values). Furthermore, the technical indicators are computed through the technical analysis of historical trading signals, which have been demonstrated to be highly effective in quantitative trading. These technical indicators include the following.

- 1) *Moving Average Indicators*: Simple moving average (SMA), exponential moving average (EMA), weighted moving average (WMA), double EMA (DEMA), triple EMA (TEMA), triangular moving average (TRIMA), Kaufman adaptive moving average (KAMA), mesa adaptive moving average (MAMA), and triple smoothed EMA (T3). These indicators smooth price signals over time to identify trends and patterns inherent in price movement.
- 2) *Momentum Indicators*: Moving average convergence/divergence (MACD), rate of change (ROC), relative strength index (RSI), stochastic oscillator (STOCH), and momentum price indicator (MPI). These indicators evaluate the strength and speed of price changes and detect potential trend reversals or continuations.
- 3) *Volatility Indicators*: Average true range (ATR), normalized ATR (NATR), and true range (TRANGE). These indicators quantify the price volatility and gauge the level of risk in the market.
- 4) *Volume Indicators*: On-balance volume (OBV), money flow index (MFI), and volume-WMA (VWMA). These indicators assess the relationship between price and trading volume and identify the accumulation or distribution of a security.

As previously mentioned, only a few stocks will be significantly influenced by relevant news. Therefore, the proposed model must not only integrate trimodal features, but also adeptly accommodate the long-tailed feature distribution.

IV. PA-TMM ARCHITECTURE

In this section, we present the PA-TMM framework as depicted in Fig. 2, where the key motivation lies in dealing with the long tail effect in feature distribution. The main structure of our model consists of two subnetworks. One is the *cross-modal fusion* module, which tackles the missing textual modality and integrates cross-modal information for automatically generating news sentiments and hybrid stock representations. The other is the *graph dual-attention* module, which dynamically infers a partial-bipartite stock attention network by considering the news- and price-induced interactions separately.

A. Cross-Modal Fusion Module

We commence by employing a cross-stream architecture to address the missing textual modality, generate sentiment prompts for other stocks, and integrate trimodal information including time series (trading signals), tabular features (technical indicators), and natural languages (textual news).

1) *Pseudo-News Padding and Activation State*: As stated before, news may be absent for certain stocks on a given day. Therefore, we fill the news position with pseudo-news, i.e., a space character, for a stock without textual modality. To differentiate pseudo-news from the real news, we divided the node set $\mathcal{V}$ into two mutually exclusive subsets on the day t : a nonactivation subset $\mathcal{V}^{(0)}$ and an activation subset $\mathcal{V}^{(1)}$. Here, $i \in \mathcal{V}^{(0)}$ indicates that stock i contains price-only information (trading signals and technical indicators), while $i \in \mathcal{V}^{(1)}$ indicates the presence of real news and thus stock i can be activated as prompts for peer stocks. By taking pseudo-news padding, we can address the issue of modality incompleteness with flexibility.

2) *Representation Learning*: This learning module maps trimodal features into representations for each stock. For the textual news, we use the pretrained language model BERT [55] as the feature extractor $\Phi^n : \mathcal{T} \rightarrow \mathbb{R}^{d_n}$ to encode the l th textual sequence $\mathbf{s}_i^{t,l} \in \mathcal{T}$ of stock $i \in \mathcal{V}$ on the day t (before market close) into a vector. Then, we take the average of all these embeddings belonging to the same stock as the news representation $\mathbf{m}_i = (1/L) \sum_{l=1}^L \Phi^n(\mathbf{s}_i^{t,l})$, where L is the number of stock-specific news on the target trading day. The knowledge contained within $\mathbf{m}_i \in \mathbb{R}^{d_n}$ is regarded as the news-induced movement.

Similarly, we encode the time-series trading signals into a vector with the bidirectional LSTM (Bi-LSTM) [37], denoted as $\Phi^p : \mathbb{R}^{T \times d_x} \rightarrow \mathbb{R}^{d_p}$, to capture the trading context of each stock, formally, $\mathbf{p}_i = \Phi^p(\mathbf{X}_i^{[t-T:t]})$, where $i \in \mathcal{V}$ and $\mathbf{X}_i^{[t-T:t]} = [\mathbf{x}_i^{t-T}, \dots, \mathbf{x}_i^{t-1}]$ denote the matrix concatenating historical trading signals. Furthermore, we use TabNet encoder [56], denoted as $\Phi^q : \mathbb{R}^{d_f} \rightarrow \mathbb{R}^{d_q}$, to map the tabular features into a continuous vector space to align with the above two modalities, that is, $\mathbf{q}_i = \Phi^q(\mathbf{I}_i^{t-1})$ and $\mathbf{I}_i^{t-1} \in \mathbb{R}^{d_f}$ denote the technical indicators of stock $i \in \mathcal{V}$ on the $(t-1)$ th day. The knowledge contained within both $\mathbf{p}_i \in \mathbb{R}^{d_p}$ and $\mathbf{q}_i \in \mathbb{R}^{d_q}$ is regarded as the price-induced movement.

3) *Modal Decomposition*: To fuse trimodal representations across news-related information and price-related information, we project them into four different spaces: two for news-stream integration and two for price-stream integration. For each stream, we have an original stream transformation for modal-specific feature extraction and a cross-stream transformation for modal-shared feature extraction

$$\begin{bmatrix} \mathbf{u}_i^m \\ \mathbf{v}_i^m \\ \mathbf{u}_i^p \\ \mathbf{v}_i^p \end{bmatrix} = \sigma \left(\begin{bmatrix} \mathbf{W}_{um} \mathbf{m}_i \\ \mathbf{W}_{vm} \mathbf{m}_i \\ \mathbf{W}_{up} [\mathbf{p}_i \| \mathbf{q}_i] \\ \mathbf{W}_{vp} [\mathbf{p}_i \| \mathbf{q}_i] \end{bmatrix} \right) \quad (2)$$

where $\sigma(\cdot)$ denotes a nonlinear activation function and “ $\|$ ” denotes the vector concatenation. $\mathbf{u}_i^m \in \mathbb{R}^{d_r}$ and $\mathbf{v}_i^m \in \mathbb{R}^{d_r}$ serve as the decomposed news-specific and news-shared representations of stock $i \in \mathcal{V}$, respectively, while $\mathbf{u}_i^p \in \mathbb{R}^{d_r}$ and $\mathbf{v}_i^p \in \mathbb{R}^{d_r}$ are the decomposed price-specific and price-shared representations, respectively. $\mathbf{W}_{um} \in \mathbb{R}^{d_r \times d_n}$, $\mathbf{W}_{vm} \in \mathbb{R}^{d_r \times d_n}$, $\mathbf{W}_{up} \in \mathbb{R}^{d_r \times (d_p + d_q)}$, and $\mathbf{W}_{vp} \in \mathbb{R}^{d_r \times (d_p + d_q)}$ are the learnable projection matrices. To ensure the independence of the decomposed modal-specific spaces from the modal-shared spaces, the orthogonal constraint can be formulated as the following orthogonal loss:

$$\mathcal{L}_{\text{ort}} = \|\mathbf{W}_{um}(\mathbf{W}_{vm})^\top\|_F + \|\mathbf{W}_{up}(\mathbf{W}_{vp})^\top\|_F \quad (3)$$

where $\|\cdot\|_F$ denotes the Forbenius norm.

4) *Modal Integration*: According to modern behavioral finance theory [6], [57], [58], investors are considered irrational and often swayed by opinions expressed in the media. Media sentiment mirrors investors' expectations regarding the future of a company or even the entire stock market, consequently contributing to stock price movements. We capture the news-driven sentiment prompts using a modal integration layer with softmax as follows:

$$\mathbf{h}_i^{\text{pmi}} = [\hat{h}_i^- \| \hat{h}_i^+] = \text{softmax}(\mathbf{W}_{sr} [\mathbf{u}_i^m \| (\mathbf{u}_i^m \odot \mathbf{v}_i^p) \| \mathbf{v}_i^p]) \quad (4)$$

where $\odot$ denotes the elementwise multiplication operation, $\mathbf{h}_i^{\text{pmi}} \in \mathbb{R}^2$ indicates the negative or positive sentiment of stock $i \in \mathcal{V}^{(1)}$ extracted from news integrating trading signals, and $\mathbf{W}_{sr} \in \mathbb{R}^{2 \times 3d_r}$ is the learned parameters. The two-category sentiment prompt readily aligns with the MPA strategy, which enhances our model's generalization performance.

Algorithm 1 Cross-Modal Fusion

Input: Node set $\mathcal{V}$;
Activated subset $\mathcal{V}^{(1)}$;
Trading day index t ;
Time series $\{\mathbf{X}_i^{[t-T:t]} | i \in \mathcal{V}\}$;
Tabular indicators $\{\mathbf{I}_i^{t-1} | i \in \mathcal{V}\}$;
Textual news corpora $\mathcal{T}$.
Output: Sentiment prompts $\{\mathbf{h}_i^{\text{pmi}} | i \in \mathcal{V}^{(1)}\}$;
Hybrid embeddings $\{\mathbf{h}_i^{\text{hyb}} | i \in \mathcal{V}\}$.

```

1 foreach  $i$  in  $\mathcal{V}$  do
2    $\mathbf{p}_i \leftarrow \text{Bi-LSTM}(\mathbf{X}_i^{[t-T:t]});$ 
3    $\mathbf{q}_i \leftarrow \text{TabNet}(\mathbf{I}_i^{t-1});$ 
4   if pre-training then
5      $\mathbf{m}_i \leftarrow \text{BERT}(\cdot);$  // pseudo-news
6      $\mathbf{u}_i^m, \mathbf{v}_i^m, \mathbf{u}_i^p, \mathbf{v}_i^p \leftarrow$ 
       modalDecomposition( $\mathbf{m}_i, \mathbf{p}_i, \mathbf{q}_i$ );
7     if  $i$  in  $\mathcal{V}^{(1)}$  then
8       Generate  $\mathbf{h}_i^{\text{pmi}}$  using EQSamp (Section V);
9     end
10     $\mathbf{h}_i^{\text{hyb}} \leftarrow \text{modalIntegration}(\mathbf{u}_i^m, \mathbf{v}_i^m, \mathbf{u}_i^p, \mathbf{v}_i^p);$ 
11  else
12    Pad the vacant news positions in  $\mathcal{T}$  with
      pseudo-news;
13     $\mathbf{m}_i \leftarrow \text{BERT}(\mathcal{T});$  // fine-tuning and testing
14     $\mathbf{u}_i^m, \mathbf{v}_i^m, \mathbf{u}_i^p, \mathbf{v}_i^p \leftarrow$ 
      modalDecomposition( $\mathbf{m}_i, \mathbf{p}_i, \mathbf{q}_i$ );
15    if  $i$  in  $\mathcal{V}^{(1)}$  then
16       $\mathbf{h}_i^{\text{pmi}} \leftarrow \text{modalIntegration}(\mathbf{u}_i^m, \mathbf{v}_i^m, \mathbf{u}_i^p, \mathbf{v}_i^p);$ 
17    end
18     $\mathbf{h}_i^{\text{hyb}} \leftarrow \text{modalIntegration}(\mathbf{u}_i^m, \mathbf{v}_i^m, \mathbf{u}_i^p, \mathbf{v}_i^p);$ 
19  end
20 end
21 return  $\{\mathbf{h}_i^{\text{pmi}} | i \in \mathcal{V}^{(1)}\}, \{\mathbf{h}_i^{\text{hyb}} | i \in \mathcal{V}\}$ 
```

Meanwhile, another modal integration layer is employed to produce the hybrid embedding $\mathbf{h}_i^{\text{hyb}} \in \mathbb{R}^{d_h}$ for stock $i \in \mathcal{V}$ from trimodal features, which serves as the stock representation

$$\mathbf{h}_i^{\text{hyb}} = \sigma(\mathbf{W}_{hr} [\mathbf{u}_i^p \| (\mathbf{u}_i^p + \mathbf{v}_i^m) \| \mathbf{v}_i^m] + \mathbf{b}_h) \quad (5)$$

where $\mathbf{W}_{hr} \in \mathbb{R}^{d_h \times 3d_r}$ and $\mathbf{b}_h \in \mathbb{R}^{d_h}$ are the learned parameters. Note that in the price-stream integration, price and shared news information are equally crucial to represent a stock; thus, the addition operation is adopted in the hybrid embedding process. However, in the news-stream integration, the news serves as the primary carrier of sentiment, and price-shared information is intended to act as a gate (multiplication operation) for filtering out noise in the news. Algorithm 1 outlines the algorithm flow for the forward propagation of the cross-modal fusion module.

B. Graph Dual-Attention Module

Stocks not only respond to news and prices, but are also susceptible to momentum spillover effects from related stocks, attributable to market anomalies [14], [15], [28]. Since the imbalance of feature distribution, direct aggregation which presumes the uniformity of node representations is deemed inappropriate. Hence, a graph dual-attention module is designed to circumvent direct aggregation to encode the exchanged heterogeneous information among stocks.

1) *Stock Polarized Activation*: While we have obtained the integrated trimodal representation (hybrid embedding) of each stock, the asymmetry feature distribution suggests that stocks with news (activated stocks) will carry more fundamental information beyond the market. This indicates that different stocks influence one another in different ways depending on their activation states. To capture this, we embed activated and nonactivated stocks separately to assist the model in distinguishing between real news and pseudo-news

$$\mathbf{n}_i = \begin{cases} \sigma(\mathbf{W}_{nh}^{(1)}[\mathbf{h}_i^{\text{hyb}} \parallel \mathbf{h}_i^{\text{pmt}}] + \mathbf{b}_n^{(1)}), & i \in \mathcal{V}^{(1)} \\ \sigma(\mathbf{W}_{nh}^{(0)}\mathbf{h}_i^{\text{hyb}} + \mathbf{b}_n^{(0)}), & i \in \mathcal{V}^{(0)} \end{cases} \quad (6)$$

where $\mathbf{n}_i$ is the node vector of stock i , and $\mathbf{W}_{nh}^{(1)} \in \mathbb{R}^{d_n \times (d_h+2)}$, $\mathbf{W}_{nh}^{(0)} \in \mathbb{R}^{d_n \times d_h}$, $\mathbf{b}_n^{(1)} \in \mathbb{R}^{d_n}$, and $\mathbf{b}_n^{(0)} \in \mathbb{R}^{d_n}$ are the learnable parameters. Since the news sentiments are only conveyed by the prompts of activated nodes, we concatenate prompts and hybrid embeddings for mapping, while for nonactivated nodes, only hybrid embeddings are fed forward.

Empirically, activated nodes with opposing sentiments are intended to be alienated, whereas those with the same sentiment are intended to be brought into closer proximity. Here, we employ cosine distance to measure the polarization loss of them

$$\mathcal{L}_{\text{pol}} = \sum_{i \in \mathcal{V}^{(1)}} \sum_{j \in \mathcal{V}^{(1)}} \cos(\mathbf{n}_i, \mathbf{n}_j) \text{sgn}((\hat{h}_i^+ - \hat{h}_i^-)(\hat{h}_j^+ - \hat{h}_j^-)) \quad (7)$$

where $\text{sgn}(\cdot)$ represents the sign function that discerns whether nodes i and j share the same sentiment. The sign of a single distance term is positive if nodes i and j share the same sentiment polarity, leading to the minimization of their distance cost; otherwise, it is negative. By minimizing the polarization loss, stocks with opposite sentiments are separated, while stocks with similar sentiments are drawn closer together.

2) *Interaction Inference*: Stocks often interact dynamically based on real-time market movements, and thus, the pre-defined, hard-coded stock networks fail to capture the full complexity of these relationships. To model these interactions, we employ a graph dual-attention mechanism that learns the attention weights, reflecting the flow of information between nodes. Specifically, for nonactivated stocks, we restrict the information exchange to occur either between activated and nonactivated stocks or among nonactivated stocks themselves. That is, the stock attention network should be partially bipartite as illustrated in Fig. 3. For a target nonactivated node $i \in \mathcal{V}^{(0)}$ and a source node $j \in \mathcal{V}^{(k)}$ where $k \in \{0, 1\}$, the normalized directed connection strength $\alpha_{i,j}^{(k)}$ is quantified by

$$\alpha_{i,j}^{(k)} = \frac{\exp(\varphi(\mathbf{n}_i, \mathbf{n}_j))}{\sum_{j \in \mathcal{V}^{(k)}} \exp(\varphi(\mathbf{n}_i, \mathbf{n}_j))} \quad (8)$$

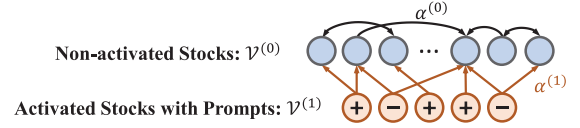


Fig. 3. Inferred partial-bipartite stock attention network.

where $\varphi(\cdot)$ estimates the message flux for every node pair. Here, we leverage the dynamic graph attention mechanism [26] that is more expressive in handling partial-bipartite graphs, denoted as

$$\varphi(\mathbf{n}_i, \mathbf{n}_j) = \mathbf{a}_\varphi^\top \text{LeakyReLU}(\mathbf{W}_{\varphi n}[\mathbf{n}_i \parallel \mathbf{n}_j]) \quad (9)$$

where the learned parameters $\mathbf{a}_\varphi \in \mathbb{R}^{d_\varphi}$ and $\mathbf{W}_{\varphi n} \in \mathbb{R}^{d_\varphi \times 2d_n}$ are applied discontinuously.

3) *Information Exchange*: In a market environment, every stock is influenced by both news- and price-driven movements. For activated stocks, however, breaking news often dominates price movements [12]. For a nonactivated stock $i \in \mathcal{V}^{(0)}$, the interactions it receives from peer stocks can be aggregated into a message vector $\tilde{\mathbf{m}}_i \in \mathbb{R}^{2d_e}$ encapsulating both types of information

$$\tilde{\mathbf{m}}_i = \left\| \sigma \left(\sum_{k \in \{0,1\}} \sum_{j \in \mathcal{V}^{(k)}} \alpha_{i,j}^{(k)} \mathbf{e}_{i,j} \right) \right\| \quad (10)$$

where $\alpha_{i,j}^{(k)}$ is the attention score governing the strength of the connection, and the edge representation $\mathbf{e}_{i,j}$ encapsulates the information flow

$$\mathbf{e}_{i,j} = \mathbf{W}_{e0}[\mathbf{n}_i \parallel \sigma(\mathbf{W}_{on}[\mathbf{n}_i \parallel \mathbf{n}_j]) \parallel \mathbf{n}_j]. \quad (11)$$

Here, learnable parameters $\mathbf{W}_{e0} \in \mathbb{R}^{d_e \times 3d_n}$ and $\mathbf{W}_{on} \in \mathbb{R}^{d_n \times 2d_n}$ are used in a skip-linked MLP, which encodes the stock-to-stock interaction into a stock-sensitive representation $\mathbf{e}_{i,j}$.

4) *Output Mapping*: Finally, for nonactivated stocks, we use a feed-forward neural network to predict the movement of stock $i \in \mathcal{V}^{(0)}$ on the next trading day. The predicted movement $\hat{\mathbf{y}}_i \in \mathbb{R}^2$ is generated as

$$\hat{\mathbf{y}}_i = [\hat{y}_i^+ \parallel \hat{y}_i^-] = \text{softmax}(\mathbf{W}_i[\mathbf{n}_i \parallel \tilde{\mathbf{m}}_i] + \mathbf{b}_i) \quad (12)$$

where $\mathbf{W}_i \in \mathbb{R}^{2 \times (d_n + 2d_e)}$ and $\mathbf{b}_i \in \mathbb{R}^2$ are the learned parameters tailored for individual nodes. For activated stock $i \in \mathcal{V}^{(1)}$, the extracted sentiment prompts $\mathbf{h}_i^{\text{pmt}} = [\hat{h}_i^- \parallel \hat{h}_i^+]$ directly correspond to its predicted price movement. Algorithm 2 outlines the algorithm flow for the forward propagation of the graph dual-attention module.

5) *Discussion*: The conventional graph attention mechanism is primarily designed for homogeneous graphs, while the method proposed in this article requires modeling the stock network as a partially bipartite graph, as illustrated in Fig. 3. Ablation experiments (Section VI-C) demonstrate that removing the message vectors (w/o Msgs.) calculated by the attention scores significantly degrades model performance. Thus, it is crucial to differentiate between the two distinct types of attention scores, $\alpha_{i,j}^{(0)}$ and $\alpha_{i,j}^{(1)}$, in the graph dual-attention module. Indeed, treating activated and nonactivated nodes separately in this module may increase computational

Algorithm 2 Graph Dual-Attention

Input: Node set $\mathcal{V}$;
Activated subset $\mathcal{V}^{(1)}$;
Non-activated subset $\mathcal{V}^{(0)}$;
Trading day index t ;
Sentiment prompts $\{\mathbf{h}_i^{pmt} | i \in \mathcal{V}^{(1)}\}$;
Hybrid embeddings $\{\mathbf{h}_i^{hyb} | i \in \mathcal{V}\}$.
Output: Predicted movements $\{\hat{\mathbf{y}}_i | i \in \mathcal{V}\}$ for day t .

```

1 foreach  $i$  in  $\mathcal{V}$  do
2   if  $i$  in  $\mathcal{V}^{(1)}$  then
3      $\mathbf{n}_i \leftarrow \text{polarizedActivation}(\mathbf{h}_i^{pmt}, \mathbf{h}_i^{hyb})$ ;
4   else
5      $\mathbf{n}_i \leftarrow \text{polarizedActivation}(\mathbf{h}_i^{hyb})$ ;
6   end
7 end
8 foreach  $i$  in  $\mathcal{V}$  do
9   foreach  $j$  in  $\mathcal{V}$  do
10    if  $j$  in  $\mathcal{V}^{(1)}$  then
11       $\alpha_{i,j}^{(1)} \leftarrow \text{interactionsInference}(\mathbf{n}_i, \mathbf{n}_j)$ ;
12    else
13       $\alpha_{i,j}^{(0)} \leftarrow \text{interactionsInference}(\mathbf{n}_i, \mathbf{n}_j)$ ;
14    end
15     $\mathbf{e}_{i,j} \leftarrow \text{edgeEmbedding}(\mathbf{n}_i, \mathbf{n}_j)$ ;
16  end
17 end
18 foreach  $i$  in  $\mathcal{V}$  do
19    $\tilde{\mathbf{m}}_i \leftarrow \text{informationExchange}(\alpha_{i,\cdot}^{(0/1)}, \mathbf{e}_{i,\cdot})$ ;
20   if  $i$  in  $\mathcal{V}^{(1)}$  then
21      $\hat{\mathbf{y}}_i \leftarrow \mathbf{h}_i^{pmt}$ ;
22   else
23      $\hat{\mathbf{y}}_i \leftarrow \text{outputMapping}(\mathbf{n}_i, \tilde{\mathbf{m}}_i)$ ;
24   end
25 end
26 return  $\{\hat{\mathbf{y}}_i | i \in \mathcal{V}\}$ 

```

complexity. However, in this study, our primary focus is on the performance improvements achieved in financial prediction. Future extensions could explore the design of more lightweight attention modules to enhance computational efficiency.

C. Computational Complexity

The computational complexity of the model can be analyzed in two parts. First, in the cross-modal fusion module, the primary computational cost arises from the recurrent component of the LSTM, which has a complexity of $O(N \times T \times d_p^2)$, where N is the number of stocks, T is the length of the time series, and d_p is the hidden size of LSTM. The modal decomposition and modal integration consist of linear layers, whose complexity is negligible compared to that of the LSTM. Second, in the graph dual-attention module, the interactions inference step involves computing the unidirectional interactions between all pairs of nodes, leading to a complexity of $O(N^2 \times d_n)$, where d_n is the dimension of the node vector. The remaining components are also made up of linear layers, and their computational cost can be ignored. Therefore, the overall model complexity is $O(N \times T \times d_p^2) + O(N^2 \times d_n)$. This analysis highlights the

need to control the scale of the stock network, as excessive node interactions can lead to increased computational costs, potentially reducing the model's efficiency.

V. MODEL OPTIMIZATION

We employ a two-stage optimization method to ensure that PA-TMM converges to respond to the news sensitively rather than to be distracted by the overwhelming head features. First, we apply an MPA with a data augmentation strategy for extensive pretraining, which executes the EQSamp to facilitate the graph dual-attention module in proactively capturing news information. Second, we fine-tune the model using real news data to steer the output toward precise stock movement predictions.

A. Model Pretraining: MPA

Typically, stocks with daily news coverage are rare, constituting no more than one-tenth of the entire stock pool on average, as mentioned in Section VI-A1. The long tail effect in feature distribution presents an obstacle for the graph dual-attention module to converge toward being sufficiently sensitive to activated nodes, as it can be easily distracted by the abundance of price features. Hence, we propose an MPA strategy for the supervised pretraining of our framework.

1) *Equivalence Resampling*: The market conditions reflected in historical data may not necessarily correspond to future market states. To tackle this, we propose the EQSamp strategy for data augmentation in stock market datasets, which not only compels the model to adapt to the long-tail effects of features but also enables it to accommodate a wide range of possible scenarios. Considering the dominant influence of news, this strategy establishes an equivalence between market sentiments in news and stock movements. Different from (4), we generate prompts directly via EQSamp in the pretraining process rather than relying on the cross-modal fusion module. We begin by randomly activating a stock subset $\mathcal{V}^{(1)} \subset \mathcal{V}$ on each trading day, and the set size is dynamically adjusted to emulate daily changes in the number of stocks that carry news. Essentially, the quantity-varying process of stocks with news is a counting process that records their number within a unit of time. We approximate this process using a Poisson process when stock sampling

$$\mathbb{P}(|\mathcal{V}^{(1)}| = k) = \frac{\lambda^k \exp(-\lambda)}{k!} \quad (13)$$

where λ is determined through maximum likelihood estimation (MLE). As a next step, we assign them movement prompts based on their ground-truth movements, which are treated as an equivalent surrogate for news sentiments

$$\mathbf{h}_i^{pmt} = [\hat{h}_i^-, \hat{h}_i^+] = \begin{cases} [(1 - \xi_i) \|\xi_i], & y_i^t = 0 \\ [\xi_i \| (1 - \xi_i)], & y_i^t = 1 \end{cases} \quad (14)$$

where $i \in \mathcal{V}^{(1)}$ and the noise term ξ_i follows a uniform distribution $\mathcal{U}(0, 0.5)$, introducing random confidence to the prompt to enhance robustness [59]. However, in the real world, the insights within news may imply ambiguous price movements; in other words, some sentiments may be misidentified in

practical sentiment inference. To prevent the model from over-fitting due to over-reliance on activated nodes, we introduce a strategy of inverting movement prompts with a mutation probability θ during pretraining, that is, $\mathbf{h}_i^{\text{pmt}} \leftarrow 1 - \mathbf{h}_i^{\text{pmt}}$. It is worth noting that MPA can perform multiple samplings to obtain numerous different activable subsets for a single day to augment the pretraining data. This capability enhances the instance reuse rate, facilitating extensive pretraining in financial scenarios and thus avoiding the biased attention in the graph attention mechanism.

2) *Pretraining Objectives*: MPA aims to avoid insufficient attention to activated stocks caused by under-fitting through substantial movement prompts that are part of the input. Since news sentiment prompts are replaced by the generated movement prompts during pretraining, we set all input news to be pseudo-news (Section IV-A1). In this way, the movement loss can be measured solely by the predicted movements of nonactivated stocks

$$\mathcal{L}_{\text{mov}}^{(0)} = -\frac{1}{|\mathcal{V}|} \left(\sum_{i \in \mathcal{V}^{(0)}} ((1 - y_i^t) \log(\hat{y}_i^-) + y_i^t \log(\hat{y}_i^+)) \right) \quad (15)$$

where y_i^t denotes the ground-truth movement of stock $i \in \mathcal{V}^{(0)}$. Then, we formulate the pretraining objective by amalgamating all loss terms with different weights

$$\mathcal{L}_P = \sum_i (\mathcal{L}_{\text{mov}}^{(0)} + \beta \mathcal{L}_{\text{ort}} + \gamma \mathcal{L}_{\text{pol}}) \quad (16)$$

where β and γ are the weighting factors. Algorithm 3 outlines the pseudocode for model pretraining.

B. Model Fine-Tuning

During pretraining, our model primarily focuses on the graph dual-attention module, and the optimization of the ability of the cross-modal fusion module to automatically extract news sentiment prompts is neglected. Therefore, we leverage all available real news data to refine the cross-modal fusion module during fine-tuning. In this stage, sentiment prompts are equivalent to the movements, whose positive/negative sentiments correspond to rise/fall. Thus, the movement loss for activated stocks can also be calculated by the cross-entropy loss

$$\mathcal{L}_{\text{mov}}^{(1)} = -\frac{1}{|\mathcal{V}|} \left(\sum_{i \in \mathcal{V}^{(1)}} ((1 - y_i^t) \log(\hat{h}_i^-) + y_i^t \log(\hat{h}_i^+)) \right) \quad (17)$$

where y_i^t denotes the ground-truth movement of stock $i \in \mathcal{V}^{(1)}$. Similarly, we formulate the fine-tuning objective

$$\mathcal{L}_F = \sum_i (\mathcal{L}_{\text{mov}}^{(1)} + \mathcal{L}_{\text{mov}}^{(0)} + \beta \mathcal{L}_{\text{ort}} + \gamma \mathcal{L}_{\text{pol}}). \quad (18)$$

Algorithm 4 outlines the pseudocode for model fine-tuning. Overall, the consistency of the optimization objectives between the pretraining and fine-tuning stages ensures a seamless and nondivergent transition between the two steps, contributing to enhancing the performance of stock movement prediction.

Algorithm 3 Model Pretraining

Input: Node set $\mathcal{V}$;
Pre-training dataset with length T_p ;
Time series $\{\mathbf{X}_i^{[t-T:t]} | i \in \mathcal{V}, T < t \leq T_p\}$;
Tabular indicators $\{\mathbf{I}_i^{t-1} | i \in \mathcal{V}, T < t \leq T_p\}$.
Result: Pre-trained PA-TMM model.

- 1 Initialize model parameters;
- 2 Shuffle and batch the pre-training dataset;
- 3 **foreach** *batch of pre-training dataset* **do**
- 4 **foreach** *t in the batch* **do**
- 5 $\mathcal{L}_{\text{ort}} \leftarrow 0, \mathcal{L}_{\text{pol}} \leftarrow 0, \mathcal{L}_{\text{mov}}^{(0)} \leftarrow 0$;
- 6 Pad all the news positions with pseudo-news;
- 7 Sample an activated subset $\mathcal{V}^{(1)}$ from $\mathcal{V}$;
- 8 Generate $\{\mathbf{h}_i^{\text{pmt}} | i \in \mathcal{V}^{(1)}\}$ using EQSamp;
- 9 // *Cross-Modal Fusion Module*
- 10 **foreach** *i in* $\mathcal{V}$ **do**
- 11 $\mathbf{h}_i^{\text{hyb}} \leftarrow \text{crossModalFusion}(\mathbf{X}_i^{[t-T:t]}, \mathbf{I}_i^{t-1})$;
- 12 **end**
- 13 Accumulate orthogonal loss to $\mathcal{L}_{\text{ort}}$;
- 14 // *Graph Dual-Attention Module*
- 15 **foreach** *i in* $\mathcal{V}$ **do**
- 16 $\hat{y}_i \leftarrow \text{graphDualAttention}(\{\mathbf{h}_i^{\text{pmt}} | i \in \mathcal{V}^{(1)}\}, \{\mathbf{h}_i^{\text{hyb}} | i \in \mathcal{V}\})$;
- 17 **end**
- 18 Accumulate polarization loss to $\mathcal{L}_{\text{pol}}$;
- 19 Accumulate movement loss to $\mathcal{L}_{\text{mov}}^{(0)}$ for non-activated stocks;
- 20 **end**
- 21 Calculate final pre-training loss $\mathcal{L}_P$ for the batch;
- 22 Backward and update model parameters;
- 23 **end**

VI. EXPERIMENTS

In this section, we perform extensive experiments to evaluate the effectiveness, profitability, and robustness of our PA-TMM, and conduct a case study on stock attention networks to validate our motivation.

A. Evaluation Setup

1) *Datasets*: We collected historical trading data for NASDAQ 100 entities and S&P 500 entities spanning from January 2014 to December 2019, sourced from Yahoo Finance² and Nasdaq Data Link.³ Note that Dow Jones Industrial Average (DJIA) consists of only 30 stocks, which may result in a network that is too small in scale to fully leverage the news information from other stocks. For this reason, we do not conduct experiments on DJIA. However, the stocks comprising the DJIA are generally included in the NASDAQ 100 and S&P 500 indices. As such, the DJIA stocks are also effectively considered in our experiments. As shown in Table I, a total of 118 stocks from the NASDAQ 100 index and 510 stocks from the S&P 500 index are incorporated into our evaluation. These datasets include each stock's fundamental historical trading information, including the highest price, lowest price,

²<https://finance.yahoo.com>

³<https://data.nasdaq.com>

Algorithm 4 Model Fine-Tuning

Input: Node set $\mathcal{V}$;
 Fine-tuning dataset with length T_f ;
 Time series $\{\mathbf{X}_i^{[t-T:t]} | i \in \mathcal{V}, T < t \leq T_f\}$;
 Tabular indicators $\{\mathbf{I}_i^{t-1} | i \in \mathcal{V}, T < t \leq T_f\}$;
 Textual news corpora $\mathcal{T}$.
Result: Fine-tuned PA-TMM model.

- 1 Load the pre-trained model parameters;
- 2 Shuffle and batch the fine-tuning dataset;
- 3 **foreach** *batch of fine-tuning dataset* **do**
- 4 **foreach** *t in the batch* **do**
- 5 $\mathcal{L}_{ort} \leftarrow 0, \mathcal{L}_{pol} \leftarrow 0, \mathcal{L}_{mov}^{(0)} \leftarrow 0, \mathcal{L}_{mov}^{(1)} \leftarrow 0$;
- 6 Pad the vacant news positions in $\mathcal{T}$ with pseudo-news;
- 7 Divide $\mathcal{V}$ into an activated subset $\mathcal{V}^{(1)}$ and a non-activated subset $\mathcal{V}^{(0)}$ according to $\mathcal{T}$;
- 8 *// Cross-Modal Fusion Module*
- 9 **foreach** *i in $\mathcal{V}$* **do**
- 10 **if** *i in $\mathcal{V}^{(1)}$* **then**
- 11 $\mathbf{h}_i^{pmt}, \mathbf{h}_i^{hyb} \leftarrow$
 crossModalFusion($\mathbf{X}_i^{[t-T:t]}, \mathbf{I}_i^{t-1}, \mathcal{T}$);
- 12 **else**
- 13 $\mathbf{h}_i^{hyb} \leftarrow$
 crossModalFusion($\mathbf{X}_i^{[t-T:t]}, \mathbf{I}_i^{t-1}, \mathcal{T}$);
- 14 **end**
- 15 **end**
- 16 Accumulate orthogonal loss to $\mathcal{L}_{ort}$;
- 17 *// Graph Dual-Attention Module*
- 18 **foreach** *i in $\mathcal{V}$* **do**
- 19 $\hat{\mathbf{y}}_i \leftarrow$ graphDualAttention(
 $\{\mathbf{h}_i^{pmt} | i \in \mathcal{V}^{(1)}\}, \{\mathbf{h}_i^{hyb} | i \in \mathcal{V}\}$);
- 20 **end**
- 21 Accumulate polarization loss to $\mathcal{L}_{pol}$;
- 22 Accumulate movement loss to $\mathcal{L}_{mov}^{(0)}$ and $\mathcal{L}_{mov}^{(1)}$ for non-activated and activated stocks;
- 23 **end**
- 24 Calculate final fine-tuning loss $\mathcal{L}_F$ for the batch;
- 25 Backward and update model parameters;
- 26 **end**

opening price, closing price, and trade volume. To account for variations in price levels, we standardize the price data for each stock. Also, we compute the technical indicators as described in Section III using TA-lib.⁴ Moreover, we collected news headlines for these entities spanning from January 2016 to December 2019 from Benzinga,⁵ which had labeled the relevant stocks impacted by each news item. A total of 10 536 news articles are included. On average, only about 11 out of 118 stocks and 26 out of 510 stocks are directly associated with specific daily news.

2) *Compared Baselines:* We perform a thorough comparative analysis of the proposed model against nine state-of-the-art baselines, which are categorized into three groups.

TABLE I
STATISTICS OF DATASETS

	NASDAQ 100	S&P 500
# Stocks (Nodes)	118	510
# Stocks with news (Average activated nodes)	11	26
Pre-training period	Jan. 2014 - Dec. 2015	Jan. 2014 - Dec. 2015
Fine-tuning and validation period	Jan. 2016 - Dec. 2018	Jan. 2016 - Dec. 2018
Test period (12 tests in total)	Jan. 2019 - Dec. 2019	Jan. 2019 - Dec. 2019

- 1) Sequential models, including LSTM [37], Transformer [39], frequency interpolation time series analysis baseline (FITS) [60], and Pathformer [61]. These methods are designed for time-series forecasting, and therefore, they make predictions solely based on historical time-series data.
- 2) Graph-based methods, including ESTIMATE [2], temporal graph convolution (TGC) [23], subsequence-based graph routing network (S-GRN) [46], and SAMBA [49]. These methods leverage trading signals and GNN variants for prediction. Specifically, ESTIMATE integrates temporal generative filters and wavelet hypergraph attention to capture stock behaviors and correlations. TGC jointly models the temporal evolution and relation network of the stock relationships for relational stock ranking. S-GRN extends GNNs with multiple messaging parameter sets and selects the optimal one to construct the most effective stock network. SAMBA integrates the Mamba architecture and GNNs to achieve near-linear computational complexity.
- 3) Multimodal methods, including PEN [40], STHAN-SR [14], AD-GAT [15], DANSMP [6], MCASP [62], and MSMF [54]. Specifically, PEN employs variational auto-encoders (VAEs) to analyze and capture the correlations between news and price series. STHAN-SR incorporates domain knowledge, including sector-industry information and hypergraphs, to model interactions between stocks. AD-GAT and DANSMP embed both historical news and time-series data and leverage GNNs to aggregate stock interactions using predefined stock relations or market knowledge graphs. MCASP constructs a self-attention network that employs multiple Transformer models to effectively capture intramodal information, focusing on learning from each modality independently before integrating them. MSMF leverages blank learning and progressive fusion to balance complementarity and redundancy across modalities. Among these models, only MCASP and MSMF originally utilize trimodal features (time series, news, and technical indicators). In contrast, the other models (PEN, AD-GAT, STHAN-SR, and DANSMP) initially consider only bimodal features (time series and news). To ensure a fair comparison with our proposed trimodal model, we extended these models into trimodal methods by concatenating technical indicators with time-series embeddings.

⁴<https://ta-lib.org>

⁵<https://github.com/Benzinga/python-client>

3) *Evaluation Metrics*: We evaluate the performance of stock movement prediction by two standard metrics: accuracy (ACC) and Mathew's correlation coefficient (MCC) [63]. ACC measures the ratio of correctly predicted labels (both positive and negative) to the total number of predictions, which is defined as

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

where TP, TN, FP, and FN denote the true positives, true negatives, false positives, and false negatives, respectively. MCC, on the other hand, is specifically designed to handle imbalanced datasets, which is defined as

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}.$$

Furthermore, we employ a simulated trading portfolio to assess the backtesting profitability of different models, considering two metrics: annualized return rate (ARR) and annualized sharp ratio (ASR). Specifically, ARR represents the cumulative performance of a trading strategy over time. It is derived by summing up the daily returns, which are typically calculated as the percentage change in the portfolio's value from one day to the next

$$\text{ARR} = \frac{E - P}{P}$$

where E is the final value of the portfolio at the end of the investment period (including both the principal and returns), and P is the initial investment (the starting principal). ASR evaluates returns against volatility, quantifying the risk-adjusted profitability, which is defined as

$$\text{ASR} = \frac{\text{ARR} - R_f}{\sigma_p}.$$

Here, σ_p denotes the annualized standard deviation of the portfolio, serving as a measure of volatility, and R_f denotes the average return rate of a risk-free asset, typically represented by government bonds.

4) *Implementation Details*: Both datasets were divided into two distinct periods: January 2014–December 2015 for pre-training and January 2016–December 2019 for fine-tuning and evaluation. During pretraining, we used EQSamp to resample the same day 50 times. Although the pretraining data span only two years, we can generate 50 different activation subsets with different prompts for each day. We employed a rolling window approach [2] to evaluate the average performance of 12 test months in 2019. For the proposed framework, a grid search was employed to determine the optimal hyperparameters and Table II presents the optimal settings. All learnable parameters were initialized using Glorot initialization [64] and trained for a maximum of 200 epochs using the AdamW optimizer. On average, each training procedure (including pretraining and fine-tuning) costs 4.7 and 7.9 h and each test day costs 0.11 and 0.32 s with one NVIDIA Titan V GPU on the NASDAQ 100 and S&P 500 datasets, respectively.

5) *Trading Portfolio*: To assess the profitability of different models, we implemented a trading strategy to simulate real-world investment, referred to as backtesting. During execution, we ranked stocks within a designated stock pool using their

TABLE II
HYPERPARAMETER SETTINGS OF PA-TMM ON TWO DATASETS

Hyper-parameters	NASDAQ 100	S&P 500
Window size T	12	16
Dimension d_m	768	768
Dimension d_r	64	128
Dimension d_φ	192	192
Dimension d_e	256	320
Dimensions d_p and d_q	128	128
Dimensions d_h and d_n	256	384
Mutation probability θ	0.25	0.25
Loss weight β	0.15	0.20
Loss weight γ	0.025	0.015
Learning rate (pre-training)	5e-4	5e-5
Learning rate (fine-tuning)	2e-4	1e-5
Batch size	8	4

TABLE III
PERFORMANCE EVALUATION OF STOCK MOVEMENT PREDICTION (MEAN $\pm$ STANDARD DEVIATION)

Methods	NASDAQ 100		S&P 500	
	ACC (%)	MCC	ACC (%)	MCC
LSTM	53.41 $\pm$ 0.34	0.065 $\pm$ 0.046	53.18 $\pm$ 0.69	0.061 $\pm$ 0.034
Transformer	52.24 $\pm$ 1.32	0.043 $\pm$ 0.028	52.40 $\pm$ 1.07	0.043 $\pm$ 0.032
FITS	53.83 $\pm$ 0.80	0.073 $\pm$ 0.022	53.07 $\pm$ 0.61	0.059 $\pm$ 0.037
Pathformer	54.48 $\pm$ 1.03	0.087 $\pm$ 0.056	53.91 $\pm$ 0.95	0.072 $\pm$ 0.031
ESTIMATE	55.77 $\pm$ 0.48	0.113 $\pm$ 0.071	53.50 $\pm$ 0.67	0.078 $\pm$ 0.061
TGC	56.86 $\pm$ 0.79	0.144 $\pm$ 0.056	54.13 $\pm$ 1.14	0.093 $\pm$ 0.043
S-GRN	57.29 $\pm$ 0.55	0.143 $\pm$ 0.049	54.88 $\pm$ 0.72	0.106 $\pm$ 0.037
SAMBA	58.09 $\pm$ 0.76	0.151 $\pm$ 0.042	55.16 $\pm$ 0.89	0.117 $\pm$ 0.054
PEN	57.91 $\pm$ 0.82	0.157 $\pm$ 0.028	55.06 $\pm$ 0.96	0.114 $\pm$ 0.045
STHAN-SR	58.48 $\pm$ 0.89	0.161 $\pm$ 0.073	55.88 $\pm$ 0.47	0.125 $\pm$ 0.080
AD-GAT	58.95 $\pm$ 1.91	0.184 $\pm$ 0.093	56.39 $\pm$ 1.52	0.133 $\pm$ 0.084
MCASP	58.44 $\pm$ 1.29	0.175 $\pm$ 0.049	57.90 $\pm$ 1.31	0.152 $\pm$ 0.072
DANSMP	59.25 $\pm$ 1.14	0.186 $\pm$ 0.042	57.14 $\pm$ 0.63	0.140 $\pm$ 0.058
MSMF	59.45 $\pm$ 0.92	0.181 $\pm$ 0.055	57.44 $\pm$ 1.08	0.148 $\pm$ 0.087
PA-TMM	60.34$\pm$0.75	0.199$\pm$0.059	59.21$\pm$1.22	0.187$\pm$0.074

movement prediction scores daily. Our approach involved purchasing a maximum of 10 of the highest-ranked stocks from the top 20, ensuring they were not already present in the portfolio. Simultaneously, an equivalent quantity of the lowest-ranked stocks was sold off. This strategy effectively controlled the turnover rate, maintaining a consistent portfolio holding of 20 stocks. In our calculations, we assumed an initial account capital of U.S. \$5 million and factored in transaction costs amounting to 0.05% when buying shares and 0.15% when selling shares.

B. Stock Movement Prediction

We report the results of stock movement prediction of the baselines and our PA-TMM in Table III. Having observed that PA-TMM exhibits a smaller advantage to some models, we further report the results of the Diebold-Mariano test [65] in Table IV, where the null hypothesis is that there is no difference in the accuracy of two competing forecasts to examine whether our method is significantly different from all baselines in predicted accuracy of rise movement. The results indicate that our model outperforms all state-of-the-art baselines on both datasets in terms of ACC and MCC. For each baseline, our model demonstrates significantly superior predictive accuracy, achieving statistical significance at a minimum of 10% level on the NASDAQ 100 dataset

TABLE IV

DIEBOLD-MARIANO TEST RESULTS BETWEEN PA-TMM AND EACH BASELINE ON TWO DATASETS

Baselines	NASDAQ 100	S&P 500
LSTM	0.0000***	0.0000***
Transformer	0.0000***	0.0000***
FITS	0.0004***	0.0031***
Pathformer	0.0000***	0.0005***
ESTIMATE	0.0004***	0.0012***
TGC	0.0035***	0.0094***
S-GRN	0.0108**	0.0364**
SAMBA	0.0329**	0.0255**
PEN	0.0278**	0.0041***
STHAN-SR	0.0793*	0.0429**
AD-GAT	0.0351**	0.0230**
MCASP	0.0394**	0.0277**
DANSMP	0.0675*	0.0223**
MSMF	0.0545*	0.0381**

Note: *, **, and *** represent statistical significance at the levels of 10%, 5%, and 1%, respectively.

and 5% level on the S&P 500 dataset. This confirms the effectiveness of adapting the model to the long tail effect in feature distribution and prompting the news sentiments to the entire stock pool. By leveraging the proposed data augmentation method, i.e., EQSamp, our method enhances the attention mechanism's sensitivity to news and thus achieves more accurate predictions.

Analysis:

- 1) Sequential models including LSTM, Transformer, FITS, and Pathformer, due to overlooking intricate interdependencies in the stock network, exhibit inferior performance compared to graph-based methods. Although these models are not affected by the scales of the stock pool, it is insufficient to support accurate movement prediction with only trading signals.
- 2) ESTIMATE, TGC, S-GRN, and SAMBA, which account for stock connections, outperform time-series models. However, due to their limited consideration of external media information, these models are constrained by efficient capital markets [66]. Moreover, on the S&P 500 dataset, the information aggregation process introduces an unavoidable influx of unnecessary noise from more peer nodes, leading to suboptimal ACC scores.
- 3) By identifying and utilizing multimedia news, PEN, STHAN-SR, AD-GAT, and DANSMP demonstrate superior performance compared to nonnews methods. Even though PEN incorporates both prices and the corresponding salient text information, it performs less effectively than other similar models due to the absence of stock connection modeling. Upon comparing the results across two datasets, we observe that these models experience varying degrees of decline in ACC performance on the S&P 500 dataset, due to the disproportionate increase in the average number of nodes with news compared to the overall node count.
- 4) MCASP and MSMF achieve better performance on the S&P 500 dataset by fully implementing trimodal feature fusion and capturing intramodal information. However,

TABLE V

PERFORMANCE OF DIFFERENT PA-TMM VARIANTS

Variants	NASDAQ 100		S&P 500	
	ACC (%)	MCC	ACC (%)	MCC
w/o Pmts.	53.37	0.0741	53.82	0.0771
w/o Msgs.	54.94	0.0916	53.19	0.0734
w/o $\mathcal{L}_{ort}$	58.76	0.1740	57.45	0.1463
w/o $\mathcal{L}_{pol}$	57.98	0.1623	57.12	0.1408
w/o MPA	56.32	0.1277	53.04	0.0595
w/o MPA-P	58.36	0.1682	56.12	0.1338
w/o MPA-M	57.65	0.1543	54.81	0.0924
PA-TMM	60.34	0.1992	59.21	0.1873

during modality fusion, they overlook the long-tailed feature distribution, leading the models to overemphasize price features and consequently underutilize the news information.

- 5) Similar to preceding news-based methods, PA-TMM is susceptible to the proportion of nodes carrying news. However, after MPA pretraining, PA-TMM showcases its ability to adapt to sparse news. These enable activated nodes to receive timely attention, effectively surmounting the long tail effect inherent in stock feature distribution. On the other hand, the S&P 500 dataset, characterized by an increased number of activatable nodes, offers nonactivated nodes access to a broader array of information sources that are hard for previous methods to capture. Consequently, PA-TMM demonstrates optimal prediction performance in terms of both ACC and MCC.

C. Ablation Study

1) *Effectiveness of the Model Architecture:* To understand the contributions of different components within our model, we design four variants of PA-TMM:

- 1) w/o Pmts., which removes all sentiment prompts $\mathbf{h}_i^{\text{pmt}}$ and only uses the hybrid embeddings for aggregating stock interactions to assess the effectiveness of incorporating financial news;
- 2) w/o Msgs., which removes all message vectors $\tilde{\mathbf{m}}_i$ in output mapping to assess the effectiveness of the graph dual-attention module;
- 3) w/o $\mathcal{L}_{ort}$, which removes the orthogonal loss term in the objective, i.e., $\mathcal{L}_{ort}$, to assess the effectiveness of modal decomposition;
- 4) w/o $\mathcal{L}_{pol}$, which removes the polarization loss term in the objective, i.e., $\mathcal{L}_{pol}$, to assess the effectiveness of stock polarized activation.

As reported in the upper section of Table V, the average results highlight the conclusive role of all components. Particularly, sentiment prompts (w/o Pmts.) and the graph aggregation mechanism (w/o Msgs.) play pivotal roles in dealing with the long tail effect in feature distribution. The absence of these modules would jeopardize the feasibility of implementing MPA, further severely impacting prediction performance. Furthermore, the implementation of modal decomposition (w/o $\mathcal{L}_{ort}$) and polarized activation (w/o $\mathcal{L}_{pol}$) proves instrumental in enhancing the efficiency of utilizing news. These loss terms

complement each other, collectively enhancing the quality of representations learned at different stages.

2) *Effectiveness of MPA*: To further explore how effective MPA is, we also carry out ablation experiments on three variants of PA-TMM:

- 1) w/o MPA, which directly optimizes the model without MPA pretraining strategy;
- 2) w/o MPA-P, which pretrains the model but the number of the activated stocks is kept constant when MPA, based on the average number of stocks with daily news, rather than using a Poisson process;
- 3) w/o MPA-M, which pretrains the model but does not apply the mutation strategy within MPA.

The lower section of Table V presents the average performance of different pretraining strategies. We observe that the inclusion or exclusion of specific components of MPA has a significant impact on the prediction performance, particularly on the S&P 500 dataset. This suggests that the implementation of MPA empowers the model to be responsive to activated nodes, allowing it to promptly capture news without over-fitting (w/o MPA). The dynamic adjustment of the number of activated nodes following a Poisson distribution allows the model to preadapt to the real-world pattern (w/o MPA-P). Additionally, the mutation strategy helps to prevent the model from over-reliance on activated nodes (w/o MPA-M). Overall, MPA demonstrates remarkable effectiveness in the context of stock pools with different scales, especially on the S&P 500 dataset, where activatable stocks are rarer and feature imbalance is more significant.

D. Backtesting Profitability

We implemented a trading strategy (Section VI-A.5) to simulate real-world investment to explore the profitability of PA-TMM. The average backtesting results of all methods are presented in Table VI, which unequivocally demonstrate that PA-TMM outperforms all baselines in real-world investment. The highest ARR score underscores its exceptional profitability and the highest ASR signifies its outstanding performance on risk mitigation.

Analysis:

- 1) It reveals a discernible pattern when combining the backtesting results with the stock movement prediction performance: the more precise a model's prediction regarding movement, the greater its potential for earning. This observation resonates harmoniously with practical investment experience.
- 2) The overall performance of ARR and ASR of various models on the S&P 500 dataset tends to be subpar compared to the NASDAQ 100 dataset. This can be ascribed to the increased complexity of risk management with numerous stock choices and the noticeable decline in the accuracy of price movement prediction.
- 3) In contrast to graph-based methods, news-based multi-modal methods demonstrate higher ASR scores, signifying a notable enhancement in the model's resilience to risks when considering news information.
- 4) Given that PA-TMM disseminates information without the constraint of predefined relationships, it is more

TABLE VI
PERFORMANCE EVALUATION OF BACKTESTING PROFITABILITY

Methods	NASDAQ 100		S&P 500	
	ARR (%)	ASR	ARR (%)	ASR
LSTM	12.57	0.952	11.36	0.973
Transformer	10.47	0.873	10.02	0.791
FITS	14.15	1.025	12.08	0.997
Pathformer	16.76	1.143	13.64	1.185
ESTIMATE	19.63	1.396	12.65	1.022
TGC	22.46	1.424	15.37	1.214
S-GRN	24.02	1.969	16.83	1.581
SAMBA	24.51	1.660	17.53	1.347
PEN	24.27	1.817	19.05	1.426
STHAN-SR	25.98	1.898	20.26	1.440
AD-GAT	27.52	2.067	21.83	1.557
MCASP	26.21	2.050	23.84	1.697
DANSMF	28.07	2.155	23.23	1.682
MSMF	27.43	2.129	24.21	1.732
PA-TMM	30.44	2.381	26.90	1.942

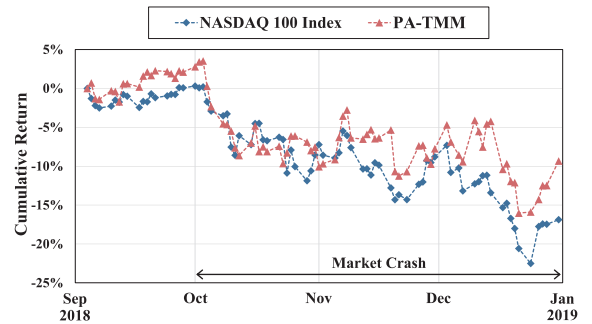


Fig. 4. Stress test during the market crash period.

effective in harnessing news sentiments. As a result, it demonstrates heightened profitability when accounting for the inherent risks involved.

E. Stress Test During Market Crash

To verify the robustness of our model against extreme market conditions, we conducted a stress test during the market crash period from October 2018 to December 2018. Fig. 4 tracks the cumulative returns of the NASDAQ 100 Index and our method. Note that the temporal length of the fine-tuning and evaluation period had to be compressed to 33 months to avoid data overlap in this test. Due to the Federal Reserve's persistent interest rate hikes in the first half of 2018, the stock market came under pressure, ultimately experiencing a widespread and highly volatile downturn. As the index encountered successive sharp declines from October, PA-TMM struggled to navigate the market's panic sentiment and even exhibited instances of erroneous judgments. However, as it progressively acclimated to the market conditions within the time window, the model showcased an aptitude for risk management. This indicates that our model demonstrates a level of resilience in extreme market conditions. Although PA-TMM falls short of generating profits during a market crash, the negative impact is still limited. In practical investment, there is often a greater emphasis on risk management capabilities than on profit-generating capabilities.

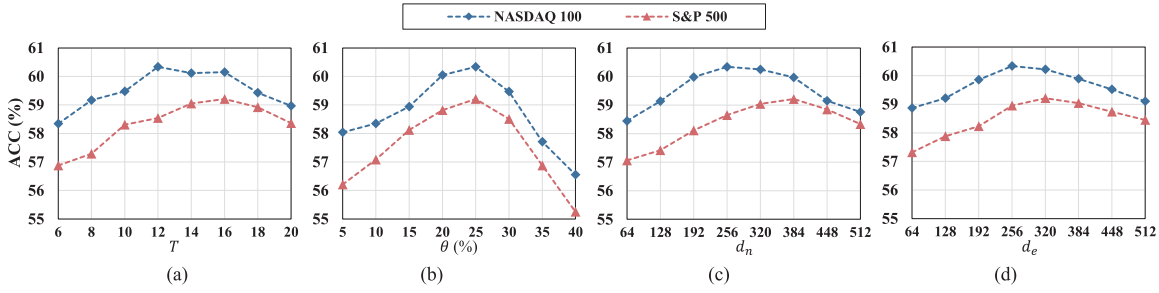


Fig. 5. Parameter sensitivity analysis with respect to the window size T , mutation probability θ , dimension d_n , and dimension d_e . (a) Window size. (b) Mutation probability. (c) Node representation. (d) Edge representation.

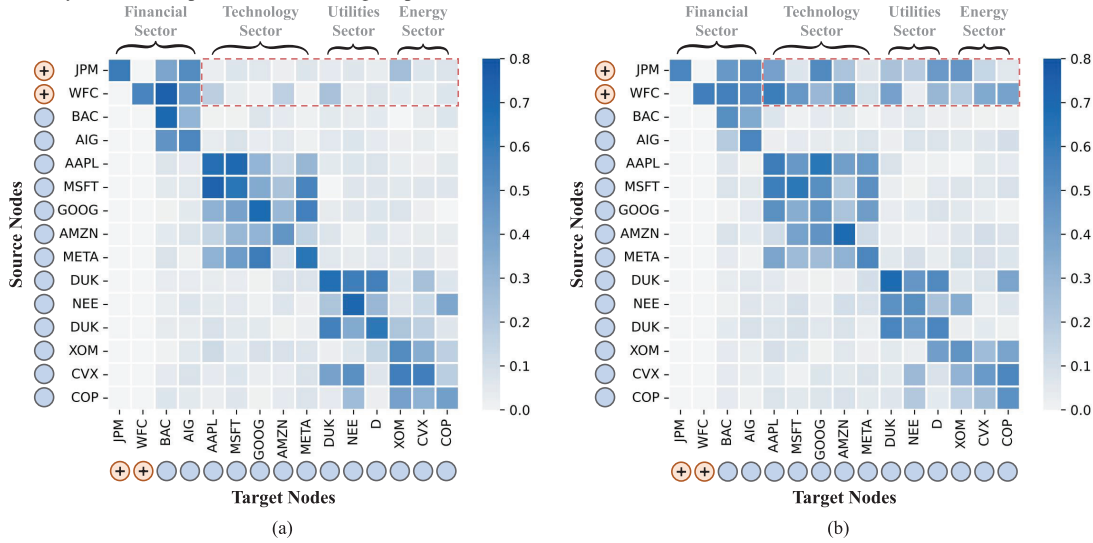


Fig. 6. Visualization of attention weights (before normalization) between some representative stocks over the S&P 500 dataset on April 12, 2019. (a) Without pretraining (w/o MPA). (b) After pretraining (PA-TMM).

F. Parameter Sensitivity Analysis

To assess the influence of different hyperparameter configurations on model performance, we investigate the sensitivity of the window size T , mutation probability θ , and dimensions d_n (node representation) and d_e (edge representation) on two datasets, as shown in Fig. 5. Overall, our model is generally stable around the optimal setting, indicating that PA-TMM is robust with different hyperparameters.

1) *Window Sizes T* : We analyze the performance variation with different window sizes in Fig. 5(a). We find that reducing the value of T causes the representation learning module to overlook long-term trends, leading to a steep decrease in the ACC score. As T increases, the larger window size may deteriorate performance by introducing stale information from past days. This suggests that choosing the appropriate window size is crucial.

2) *Mutation Probability θ* : To examine how the mutation probability in MPA influences the final performance, we report the ACC score (after fine-tuning) of varying θ in Fig. 5(b). The results indicate that the huge mutation probability leads the model to mistrust the provided movement prompts, ultimately resulting in the fine-tuned model not fully leveraging sentiments within the news to a certain extent. On the other hand, introducing appropriate mutations to prompts during MPA can effectively mitigate over-fitting and reach optimal performance after fine-tuning.

3) *Dimensions of d_n and d_e* : Fig. 5(c) and (d) demonstrates that excessively low or high dimensionality worsens the ACC score due to under-fitting and over-fitting, respectively. Hence, we suggest using the reported parameter configurations in Table II for optimization and prediction.

G. Case Study on Exploring Stock Attention Networks

We conducted a case study over the S&P500 dataset on April 12, 2019, to further investigate the effectiveness of MPA. On that day, two major U.S. banks, JPMorgan Chase (JPM) and Wells Fargo (WFC), shared their first-quarter results and the market reacted positively.⁶ This news set a positive tone for earnings overall and the two stocks imposed intense bullish sentiment in the stock network. To explore whether MPA resolves the biased attention issue caused by the long-tailed feature distribution, we visualize the attention weights of some representative stocks from financial, technology, utilities, and energy sectors in Fig. 6. In the absence of MPA [Fig. 6(a)], the attention is primarily distributed within each independent sector. Although JPM and WFC carry useful news intelligence, other stocks hardly pay attention to their strong bullish sentiments apart from those in the financial sector. This biased attention phenomenon further leads to ineffective information exchange of news in the stock network. However, after pretraining with MPA [Fig. 6(b)], there is a

⁶<https://www.benzinga.com/node/13529410>

significant increase in cross-sector attention to news carriers, as evidenced by the noticeable rise in the number of target nodes effectively influenced by JPM and WFC. Therefore, PA-TMM's capability to respond sensitively to news becomes a key impulse in enhancing performance for stock movement prediction. This advancement can be attributed to the data augmentation via EQSamp and the utilization of polarized activation for learning node representations.

VII. CONCLUSION

In this article, we propose a PA-TMM for stock movement prediction and quantitative trading, which addresses the long tail effect in financial feature distributions. We introduce a pretraining method, MPA, to optimize the model's sensitivity to news information. To tackle feature scarcity, we apply an EQSamp strategy for data augmentation during MPA. By treating stocks with news sentiment as prompts for peer stocks, PA-TMM efficiently captures cross-modal information. The experimental results show that PA-TMM outperforms the state-of-the-art baselines in stock movement prediction, with significant improvements in ACC and MCC across two real-world datasets. Further, ablation studies, case studies, backtesting, stress tests, and sensitivity analyses confirm PA-TMM's effectiveness, profitability, and robustness under varying conditions.

However, PA-TMM faces computational challenges due to the complexity of the graph dual-attention module. Future work could focus on developing a more lightweight attention mechanism. Additionally, PA-TMM heavily relies on news data, and its effectiveness may degrade when news is absent on specific trading days. Future research could address this by incorporating more diverse textual information, such as company financial reports, social media, geopolitical events, and policy regulations. These varied sources would introduce new challenges in multimodal feature fusion, but we believe they could significantly enhance the model's understanding of stock markets and improve prediction accuracy.

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Haotian Liu received the B.S. degree in automation science and engineering from Xi'an Jiaotong University, Xi'an, China, in 2022, where he is currently pursuing the M.S. degree.

His research interests include graph learning, data mining, and relevant applications.



Bowen Hu (Graduate Student Member, IEEE) received the B.S. degree in automation science and engineering from Xi'an Jiaotong University, Xi'an, China, in 2019, where he is currently pursuing the Ph.D. degree.

His research interests include data analysis and mining.



Yadong Zhou (Member, IEEE) received the B.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 2004 and 2011, respectively.

He was a Post-Doctoral Researcher with The Chinese University of Hong Kong, Hong Kong, from 2014 to 2015. He is currently a Professor with the Faculty of Electronic and Information Engineering, Xi'an Jiaotong University. He has published more than 60 articles in prestigious refereed conferences and journals, such as IEEE TRANSACTIONS ON SMART GRID, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS, *ACM Transactions on Knowledge Discovery from Data*, IEEE INFOCOM, and AAAI. His research interests include data analysis and mining, network science and its applications, and software-defined networks.



Yuxun Zhou received the B.S. degree in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 2009, the Diplôme d'Ingénieur degree in applied mathematics from École Centrale Paris, Gif-sur-Yvette, France, in 2012, and the Ph.D. degree in electrical engineering from the Department of Electrical Engineering and Computer Sciences, University of California at Berkeley, Berkeley, CA, USA, in 2017.

His research interests include artificial intelligence, statistical learning theory, and paradigms for modern information-rich, large-scale, and human-involved systems.